

FCAI ANU Materials – Semi Seniors – Final 2026 – PPIS – Dr. Sara

Part 1: Windows Forms & C# Basics

1. Which method is used to programmatically add an item to a ListBox?

- A) `ListBox.AddItem()`
- **B) `ListBox.Items.Add()`**
- C) `ListBox.Append()`
- D) `ListBox.Items.Insert()`

Explanation: In Windows Forms, the `ListBox` control has an `Items` property, which is a collection. To add an item to the end of this collection, you call the `.Add()` method on that property.

2. Which of the following is the correct syntax for `TryParse` on a string value "45.67" to parse as a double?

- A) `double.TryParse(out result, "45.67")`
- B) `double.Parse("45.67", result)`
- C) `TryParse("45.67", result)`
- **D) `double.TryParse("45.67", out result)`**

Explanation: The `TryParse` method requires the input string as the first argument and the variable where the result will be stored (prefixed with the `out` keyword) as the second argument.

3. Which of the following methods would you use to add new lines to a file without overwriting its contents?

- **A) `File.AppendText`**
- B) `File.CreateText`
- C) `File.WriteAllText`
- D) `File.ReadAllText`

Explanation: `AppendText` opens an existing file (or creates a new one) and places the cursor at the end so new data doesn't delete existing data. `WriteAllText` would overwrite the entire file.

4. To close an application's form in code, use the statement `this.Close();`

- **A) True**
- B) False

Explanation: In C#, this refers to the current instance of the `Form` class. The `.Close()` method is the standard way to shut down and dispose of that form.

5. Which property controls how the background image is displayed on the form?

- A) ImageLayout
- B) SizeMode
- **C) BackgroundImageLayout**
- D) Image

Explanation: While a PictureBox uses SizeMode, a Form uses BackgroundImageLayout to determine if the background image is tiled, centered, stretched, or zoomed.

6. What will be the output of the following code?

```
string str = "Hello"; MessageBox.Show(str.IndexOf('h').ToString());
```

- A) 0
- B) 1
- **C) -1**
- D) Error

Explanation: C# is case-sensitive. The string contains an uppercase 'H' at index 0. Because there is no lowercase 'h' in "Hello", the IndexOf method returns -1.

7. The Load event takes place after the form is displayed on the screen.

- A) True
- **B) False**

Explanation: The Load event occurs *before* a form is displayed for the first time. If you want code to run *after* the form is visible, you would use the Shown event.

8. How can you programmatically set focus to a TextBox control named textBox1?

- A) textBox1.SetFocus()
- **B) textBox1.Focus()**
- C) SetFocus(textBox1)
- D) textBox1.Focus=true

Explanation: .Focus() is a method available to all Windows Forms controls to move the user's cursor or input focus to that specific control.

9. When you call a string object's Split method, the method divides the string into substrings and returns them as an array of strings.

- **A) True**
- B) False

Explanation: This is the primary function of the Split method; it breaks a string based on a delimiter (like a comma or space) and stores the results in a string[] array.

10. Which property of the GroupBox is used to set the title displayed at the top of the box?

- **A) Text**
- B) Title
- C) Header
- D) Name

Explanation: In Windows Forms, the caption or label shown on most controls (Buttons, Labels, GroupBoxes) is stored in the Text property.

11. Which property controls how the image is sized or stretched within the PictureBox?

- A) AutoSize
- B) ImageMode
- C) Stretch
- **D) SizeMode**

Explanation: The PictureBox.SizeMode property allows you to choose options like StretchImage, Zoom, or CenterImage.

12. EndOfFile property is used to checks whether the end of the file has been reached?

- A) True
- **B) False**

Explanation: In C# StreamReader, the property is actually called EndOfStream. There is no standard EndOfFile property in the System.IO namespace.

13. What is returned when the user clicks the Open button in OpenFileDialog?

- A) Dialog.Open
- B) Dialog.OK
- C) DialogResult.Open
- **D) DialogResult.OK**

Explanation: Windows dialogs return a value from the DialogResult enumeration. If the user successfully clicks the "Open" or "OK" button, it returns DialogResult.OK.

Part 2: Advanced Concepts & Data Handling

14. The file format is commonly used to export spreadsheet data to a text file.

- A) SDV
- **B) CSV**
- C) XML
- D) PDF

Explanation: CSV (Comma Separated Values) is the industry standard for plain-text representations of spreadsheet tables.

15. In a Windows Forms application, where should you declare a field variable if you need to access it from multiple event handlers (e.g., button clicks)?

- A) Inside each method where it is used
- B) As a parameter in each method
- **C) At the class level, outside any methods**
- D) In the Main method

Explanation: Declaring a variable at the class level (a "field") gives it "class scope," meaning all methods within that form can see and modify it.

16. Forms and most controls have a ForeColor property that allows you to change the object's background color.

- A) True
- **B) False**

Explanation: ForeColor changes the color of the text (foreground). To change the background color, you must use the BackColor property.

17. rand.Next(10) returns

- **A) 0 to 9**
- B) 1 to 10
- C) 0 to 10
- D) -10 to 10

Explanation: The Random.Next(int maxValue) method returns a non-negative random integer that is less than the specified maximum. Therefore, 10 is excluded.

18. String.Compare() returns a Boolean value.

- A) True
- **B) False**

Explanation: String.Compare returns an **integer: 0** if strings are equal, a **negative** value if the first is "less than" the second, and a **positive** value otherwise.

19. What will be the output of the following code?

```
double x = 123.41; MessageBox.Show(x.ToString("n1"));
```

- A) 123.41
- **B) 123.4**
- C) \$123.41
- D) \$123.4

Explanation: The format string "n1" stands for "Numeric" with **1** decimal place. It rounds/formats the number 123.41 to 123.4.

20. Trim() removes whitespace in the middle of a string.

- A) True
- **B) False**

Explanation: Trim() only removes whitespace from the very beginning and very end of a string.

21. What will be the output of the following code?

```
string str1 = "Hello"; string str2 = str1.Remove(1); MessageBox.Show(str2);
```

- **A) H**
- B) ello
- C) Hllo
- D) e

Explanation: Remove(1) deletes all characters from index 1 to the end of the string. Since index 0 is 'H', and index 1 is 'e', everything starting from 'e' is removed, leaving only "H".

22. You add your own code to the Program.cs file as you develop an application.

- A) True
- **B) False**

Explanation: Program.cs usually only contains the Main method to start the application. Logic and UI code are placed in Form.cs files or separate class files.

23. What would you use to determine if any RadioButton in a group is selected?

- **A) if (radioButton.Checked == true)**
- B) if (radioButton.Text == "Selected")
- C) radioButton.IsChecked
- D) radioButton.Selected == true

Explanation: The property used to check the state of a RadioButton or CheckBox is .Checked.

24. Which of the following is NOT a valid value for the BorderStyle property of a Label control?

- A) FixedSingle
- B) Fixed3D
- **C) FixedDouble**
- D) None

Explanation: The BorderStyle enumeration for Labels only includes None, FixedSingle, and Fixed3D.

25. The default value of the AutoSize property of a Label control is True.

- **A) True**
- B) False

Explanation: In modern Visual Studio versions, Labels default to AutoSize = true so the control grows to fit the text you type.

26. Which property hides the characters typed in a TextBox (for passwords)?

- A) Hidden
- B) UseSystemPasswordChar
- C) Password
- **D) PasswordChar**

Explanation: While UseSystemPasswordChar is a valid boolean, PasswordChar is the property used to define which specific character (like *) is shown instead of the text. (Note: Many textbooks accept both, but PasswordChar is the classic property).

27. You can clear the contents of a TextBox control in the same way that you clear the contents of a Label control.

- **A) True**
- B) False

Explanation: Both controls use the .Text property. Setting textBox1.Text = ""; or label1.Text = ""; works for both.

Part 3: Logic & Arrays

28. Which of the following statements allows you to select the item at index 2 in a ListBox?

- A) `ListBox.SelectItem(2);`
- B) `ListBox.Items[2].Selected = true;`
- C) `ListBox.Select(2);`
- **D) `ListBox.SelectedIndex = 2;`**

Explanation: `SelectedIndex` is the standard property used to get or set the currently selected item by its numerical position.

29. Consider the code for `checkBox1_CheckedChanged`. How many times is the message shown when the user checks and then unchecks the `CheckBox1`?

- A) 0
- B) 1
- **C) 2**
- D) None of these

Explanation: The `CheckedChanged` event fires every time the `Checked` property changes. Checking it is one change (False to True); unchecking it is a second change (True to False).

30. Multiple `RadioButton` controls in the same `GroupBox` can be selected at the same time.

- A) True
- **B) False**

Explanation: `RadioButtons` are designed to be mutually exclusive within their container. Selecting one automatically deselects the others in that same `GroupBox`.

31. What is the output of the following code?

```
static void Test(int a, int b = 5, int c = 3) { Console.WriteLine(a + b + c); } ... Test(2, 3);
```

- A) 5
- **B) 8**
- C) 10
- D) Error

Explanation: The call `Test(2, 3)` assigns **2** to `a` and **3** to `b` (overriding its default). `c` uses its default value of **3**. Sum: $2 + 3 + 3 = 8$.

32. Which of the following is NOT considered an overload of the method `void Sum(int number1, int number2)`?

- A) `void Sum(int number1)`
- **B) `int Sum(int number1, int number2)`**
- C) `void Sum(double number1, int number2)`
- D) `void Sum(double number1, double number2)`

Explanation: In C#, overloads must have different parameter lists (different types or number of arguments). Changing **only** the return type (from `void` to `int`) is not allowed as an overload.

33. Which of the following is NOT the correct way to define and initialize an array of 3 integers?

- A) `int[] a = {5, 3, 4};`
- B) `int[] a; a = new int[3]{5, 3, 4};`
- C) `int[] a; a = new int[3]; a[0] = 5; a[1] = 3; a[2] = 4;`
- **D) `int[] a; a = new int{5, 3, 4};`**

Explanation: Choice D is missing the brackets `[]` after the keyword `int`. The correct syntax for initialization is `new int[] { ... }`.

34. What is the output of the `foreach` loop code?

- **Answer: Compile Error.**
 - **Why?** In the image, the code tries to do `i++` inside a `foreach` loop. In C#, the iteration variable in a `foreach` loop is **read-only**. You cannot modify it directly.
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Wish you all the best!

Fathy.